DIGITAL TWIN CHALLENGE 2026 · HAPPIEST HEALTH

Sanjeevani Twin

A hybrid physiology + AI digital twin that predicts Type 2 Diabetes glucose spikes 2 hours before they happen, explains why, and tests the fix on a virtual patient.

FHIR R4 EHR + CGM, HR, HRV, sleep, steps | Personalised physiology | Conformal AI | 3D clinician dashboard | Built for India

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THE PROBLEM

Indian diabetes care is reactive. Spikes happen between visits.

- **~101 million** Indians live with diabetes and **~136 million** with prediabetes (ICMR-INDIAB, *Lancet D&E* 2023).
- HbA1c is checked every 3 months. Daily post-meal spikes, which drive complications, go unseen.
- Indian physiology and lifestyle break Western models: the **"thin-fat" phenotype**, rice- and roti-heavy meals, late dinners, and **festival fasting** such as Navratri sabudana.
- Wearables and CGMs now generate streams of data, but doctors get **numbers, not foresight**.

> ”What will this patient’s glucose do in the next two hours, why, and what is the smallest change that prevents the spike?”

Sanjeevani answers this for every patient, continuously, from data a clinic already has or can get cheaply.

OUR APPROACH

A real twin, not just a predictor: physics + AI

EHR (static)
FHIR R4: labs, diagnoses, meds, TCF7L2 genotype, PRS

+

Wearables (dynamic)
CGM · HR · HRV · sleep stages · steps · meal and med logs

→

Personal physiology
ODE model of glucose, insulin, gut, exercise, circadian, sleep, stress. Calibrated per patient, synced every 5 min

→

AI on top
Learns what physics misses. Conformal intervals and spike risk

Predict

Glucose at 30/60/120 min with 80% intervals. Spike-above-180 risk and countdown.

Explain

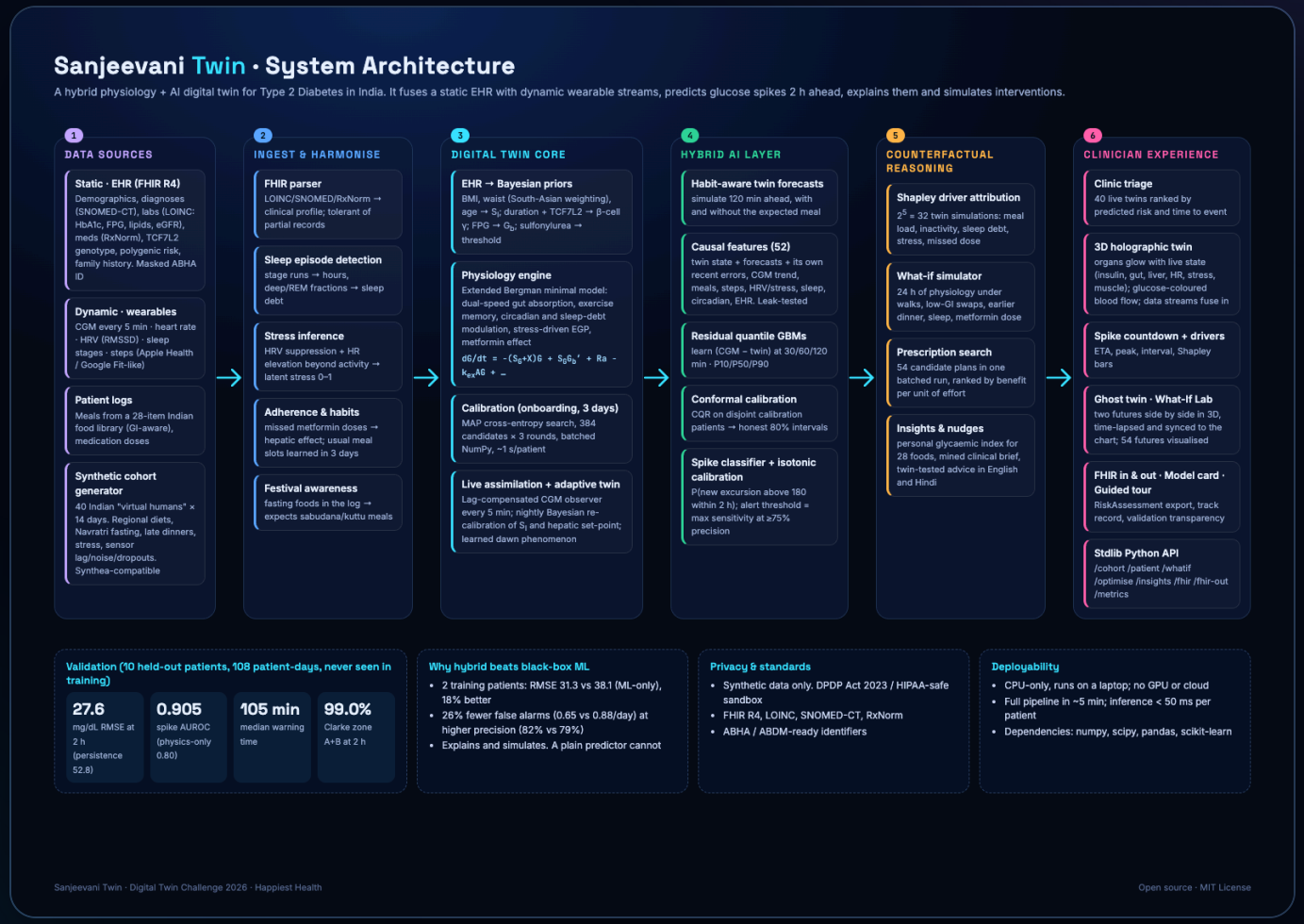
Exact Shapley attribution over 32 counterfactual twin runs: meal, inactivity, sleep debt, stress, missed dose.

Simulate

The doctor prescribes a plan and the twin simulates tomorrow. 54 candidate plans are searched in one batched run.

ARCHITECTURE

Six layers, CPU-only, four Python dependencies



DATA FUSION UNDER SANDBOX RULES

40 synthetic Indian patients × 14 days, deliberately imperfect

What we generate

- FHIR R4 bundles with LOINC/SNOMED/RxNorm codes (Synthea-compatible ingest)
- 4 regional diets and a 28-food Indian library with glycaemic-index splits
- Navratri 2026 fasting, weekend biryani, late IT-shift dinners
- CGM (5 min), HR, HRV, sleep stages, steps; meal and medication logs
- 6 story personas. All demo patients are **held out** from training

Why the evaluation is honest

- The ground truth includes effects **the twin cannot see**: dawn phenomenon, unlogged snacks, ±22% carb-count error, day-to-day insulin-sensitivity drift, latent stress
- CGM has 10-min lag, AR(1) noise, drift and dropouts
- Patient-level split: 24 train, 6 calibration, 10 test
- No real patient data. DPDP Act 2023 and HIPAA-safe

PERSONALISATION

The twin learns each patient’s hidden physiology

- **EHR → Bayesian priors.** BMI and waist (South-Asian weighting) give insulin sensitivity. Diabetes duration and **TCF7L2 rs7903146** risk alleles give beta-cell function. Fasting glucose gives the hepatic set-point.
- **3-day onboarding.** A MAP cross-entropy search fits the twin to CGM in about 1 s per patient.
- **Live sync.** A lag-compensated CGM observer keeps the twin's state current.

RECOVERED PARAMETER	R PRIOR	R TWIN
Insulin sensitivity S_I	0.76	0.89
Beta-cell responsiveness γ	0.65	0.83
Basal glucose G_b	0.86	0.98

Pearson correlation with the hidden ground truth across 40 patients. Median onboarding fit 17.3 mg/dL; open-loop simulation error 21.9 mg/dL over 11 unseen days. The twin also learns each patient's dawn phenomenon and re-calibrates itself every night (adaptive twin).

RESULTS · 10 HELD-OUT PATIENTS · 108 PATIENT-DAYS

Forecasts that beat every baseline

27.6
mg/dL RMSE at 2 h, **48% lower** than persistence

0.905
AUROC: new spike above 180 within 2 h · calibrated (ECE 0.06)

105 min
median warning time before a real excursion

99.0%
Clarke error grid zone A+B at 2 h

MODEL	RMSE 30 MIN	60 MIN	120 MIN	SPIKE AUROC	FALSE ALERTS/DAY
Persistence	26.6	42.6	52.8	—	—
Linear trend	31.2	62.3	101.3	0.546	—
Physics twin only	16.8	29.6	34.4	0.801	0.85
ML only (no twin)	15.0	24.1	28.1	0.896	0.88
Sanjeevani hybrid	**13.7**	**22.9**	**27.6**	**0.905**	**0.65**

WHY HYBRID

Physics makes AI data-efficient, quieter and explainable

−18%

error with only 2 training patients

2-h RMSE 31.3 vs 38.1 mg/dL for ML-only. The physics twin alone scores 34.4 with zero training patients, so it can start at a new clinic on day one.

−26%

false alarms

0.65 vs 0.88 per patient-day under the same alert policy, with higher precision (82% vs 79%). Alert fatigue is what kills clinical tools.

#1

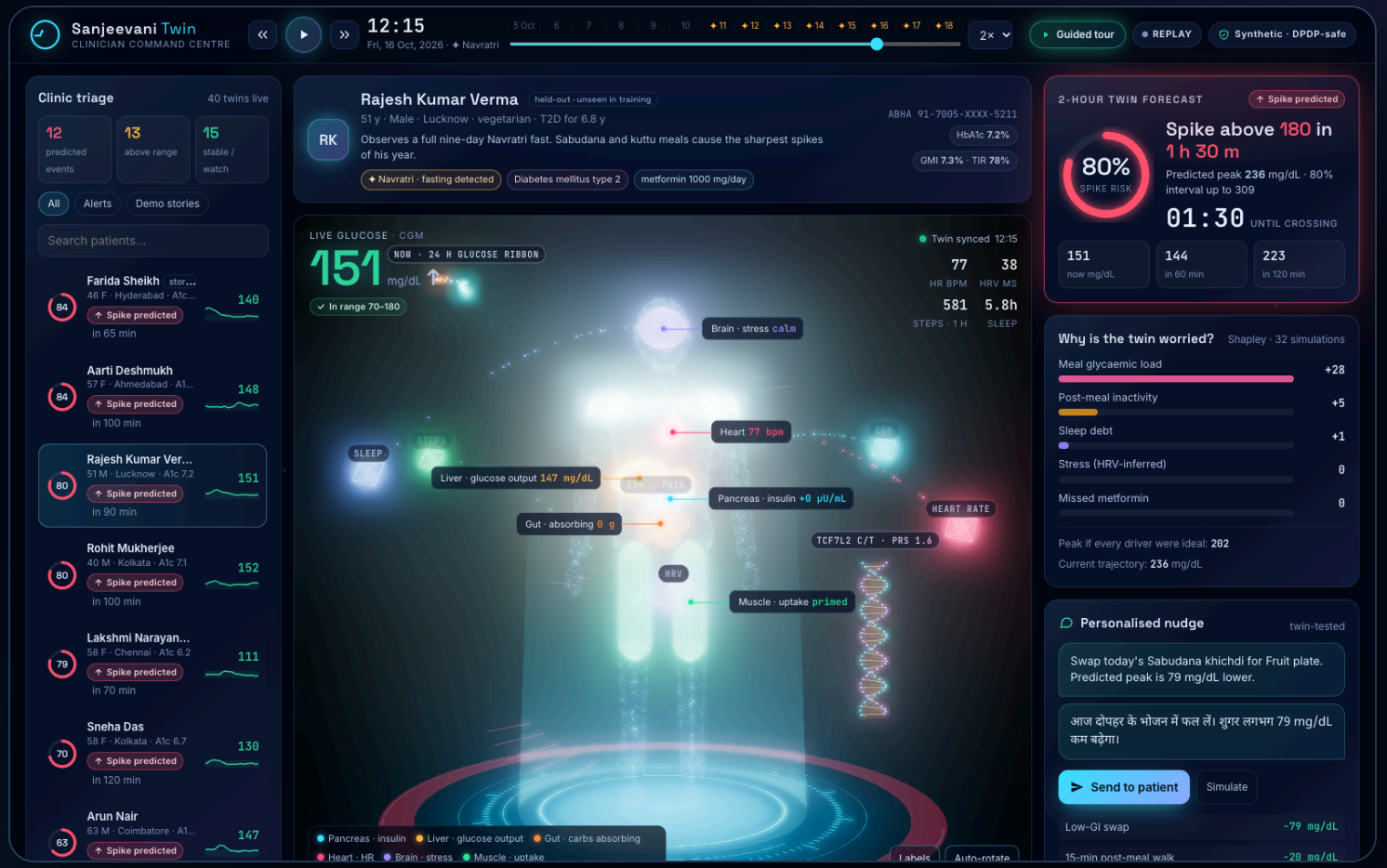
feature group = the twin

Permutation importance: removing the twin's forecasts drops AUROC by 0.154, about 8× more than any wearable group alone.

Prediction intervals use conformalized quantile regression on disjoint calibration patients (76–78% observed coverage for a nominal 80%). Spike probabilities are isotonic-calibrated. A unit test proves features never use data from after the prediction time.

CLINICIAN DASHBOARD

A living, holographic twin for every patient



STORY · RAJESH, 51, LUCKNOW · NAVRATRI, 16 OCT 2026, 12:15

The twin sees the sabudana spike coming



- Fasting foods in today's log, so the **festival-aware twin** expects a sabudana lunch
- **80% spike risk**: crossing 180 in **1 h 30 m**, predicted peak 236
- **Why:** meal glycaemic load +28 mg/dL, post-meal inactivity +5
- **Nudge, twin-tested, sent in Hindi:**
  "आज दोपहर के भोजन में फल लें। शुगर लगभग 79 mg/dL कम बढ़ेगा।"
- The actual CGM peaked at 285. The warning was real.

WHAT-IF LAB

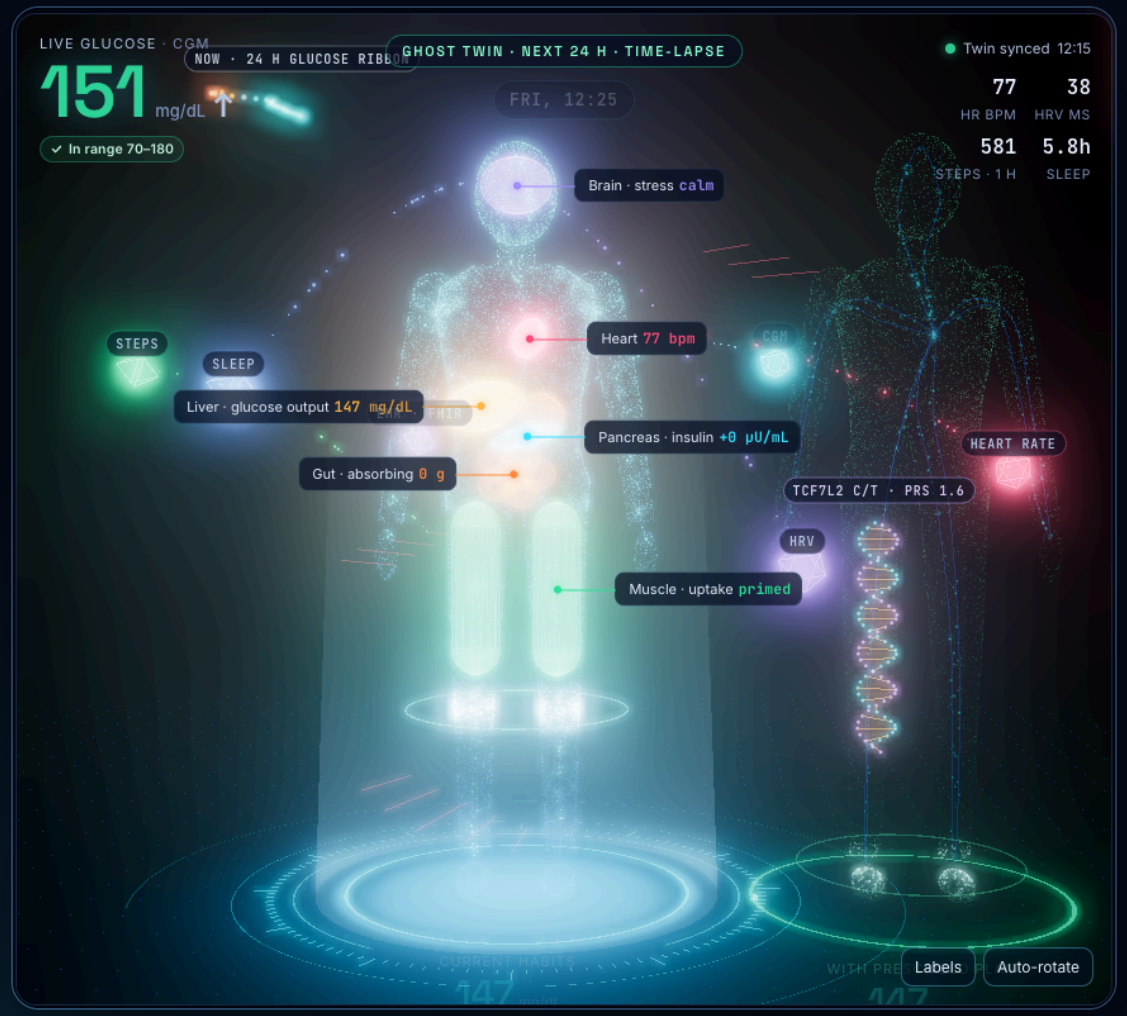
Prescribe a simulation and see tomorrow



- Sliders: post-meal walk, low-GI swap (millets, brown rice), earlier dinner, sleep, metformin dose
- **"Find the best plan"** simulates **54 possible tomorrows** in one batched run and ranks them by benefit per unit of effort
- Rajesh's recommended plan: a 20-min walk after lunch and dinner plus a low-GI swap. Peak drops by 96 mg/dL, hyperglycaemia by 72%, and projected HbA1c falls 7.2% → 7.1% in 90 days

GHOST TWIN

Two futures, side by side, before the patient changes anything



- In the What-If Lab the 3D view splits in two: **current habits** on the left and the **prescribed plan** on the right
- Both bodies are time-lapsed through the next 24 h of simulated physiology. Blood colour, glucose numbers and the chart cursor move in sync
- At 14:20 on Navratri: **273 mg/dL** on current habits vs **212 mg/dL** with the plan
- Doctors and patients *see* the benefit. That is what drives behaviour change

AI INSIGHTS

A personal glycaemic index and a clinical brief



- **Personal glycaemic index:** all 28 Indian foods simulated on *this* patient's adaptive twin, not a population GI table
- **Clinical brief** mined from 14 days of data: biggest-spike foods, fasting effect (+19 mg/dL), adherence (83%), dawn phenomenon (+32 mg/dL)
- **Track record:** the twin's own 30/60/120-min errors and interval coverage, shown openly
- **FHIR out:** predictions exported as RiskAssessment + Observation + Communication, ABDM-ready

IMPACT & DEPLOYMENT IN INDIA

Built for the PHC, not just the cloud

Clinicians

Triage 40 patients at a glance, ranked by predicted risk and time to event. Spend minutes where they matter.

Patients

Small, specific, twin-tested nudges in Hindi and English, tuned for thalis, festivals and late dinners.

Health system

FHIR R4 and ABHA-ready for the Ayushman Bharat Digital Mission. Runs on CPU with no GPU, so it suits low-resource settings.

Roadmap

- Retrospective validation on open CGM datasets (e.g. PhysioNet BIG IDEAs)
- Silent-mode pilot with a partner diabetes clinic
- More twins: hypertension and cardiovascular risk on the same engine
- Regional languages and voice nudges via ASHA workers

Responsible AI

- Synthetic data only; DPDP Act and HIPAA sandbox
- Calibrated uncertainty and a transparent model card
- Decision support only. Not a device, never doses insulin
- Open source (MIT) and fully reproducible with one command

Prediction is useful.
Simulation changes care.

Sanjeevani Twin: see the spike, know the cause, test the fix.

`github.com/batmandevx/digital-twin · python run.py`